

“PAPER_{0:D3}” -f [int]# PAPER #151 — UQFF Pillars of Creation and Rings of Relativity: MUGE Cascade Gravity Sequence

Title: UQFF Star-Magic Pillars of Creation and Rings of Relativity Gravitational Lens — MUGE 12-Term Cascade Sequence: $g=2.001e26$ and $g=5.005e25$ m/s²

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\kappa=0.0005/\text{day}$, $[SSq]=0.57$, $f_{TRZ}=0.1$)

Date: March 2026

Domain: §2.2 MUGE Compression Cycle 3 (07b7f7a6)

Source Thread: `grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt`

UQFF Mode: Superconductive Resonance (fluid cascade)

Validator: `CondensedPhysics2.py v2.1.0`, SOURCE4 (pillars_SOURCE4, rings_SOURCE4)

Cross-links: PAPER_150 (Tapestry/Westerlund, higher- g SFR regime), PAPER_152 (cosmological)

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \left(1 - [SSq] \cdot U_{bi} / F_U(r, t)\right), \quad [SSq] = 0.57$$

Abstract

The Pillars of Creation (Eagle Nebula M16 molecular pillars) and the Rings of Relativity (Einstein-ring class gravitational lens) represent two distinct astrophysical environments in which the UQFF MUGE cascade sequence reaches lower-energy configurations. Under the MUGE 12-Term Resonance framework, the Pillars yield $g = 2.001 \times 10^{26}$ m/s² and the Rings yield $g = 5.005 \times 10^{25}$ m/s² — each approximately factor 4 lower than the previous system in the 7-system cascade sequence (Tapestry/Westerlund at $1.001e27$, Pillars at $2.001e26$, Rings at $5.005e25$). This factor ~ 4 -5 cascade step represents the hierarchical de-amplification of $a_{\text{fluid_freq}}$ as system B-field and SCm density decrease from extreme SFR to supercooled molecular pillar to gravitational-lens geometry. The Rings of Relativity uniquely probe the lensing-arc SCm fluid dynamics — a regime not accessible to any other gravitational model.

1. Physical Systems

1.1 Pillars of Creation — Eagle Nebula M16

Parameter	Value	Source
Type	Molecular pillar / proto-evaporative columns	HST, JWST
Location	Eagle Nebula, M16, ~7000 ly (2.15 kpc)	VLBI
Pillar B-field	~100 muG (dense core)	Faraday rotation
Pillar density	$n_H \sim 10^{3-10} \text{ cm}^{-3}$	CO, HCO+ mapping
Column height	~4 ly (each major pillar)	HST direct imaging
Evaporation rate	~70 M_Earth/yr (EUV photoevaporation)	Hester et al. 1996
Embedded YSOs	~7 confirmed (JWST 2022)	JWST NIRCам
Age	~2-6 Myr since M16 OB stars formed	Stellar ages

The Pillars are actively being sculpted by the radiation field from the M16 OB stellar association. The SCm fluid is driven by a combination of: - EUV photoionization (from O-stars M16, ~2 kpc) - Embedded YSO outflows (from the 7+ embedded proto-stars) - Magnetic field support against gravitational collapse

1.2 Rings of Relativity — Gravitational Lens

Parameter	Value	Source
Type	Einstein ring (gravitational lens — generic class)	HST strong lensing
Lens galaxy type	Elliptical or spiral lens galaxy	Various
Einstein radius	~1-5 arcsec (typical)	Lens models
Source galaxy	High-z galaxy ($z_{\text{source}} \sim 1-3$)	Spectroscopy
Lens galaxy Geometry	$z_{\text{lens}} \sim 0.1-0.5$ Near-perfect alignment (lens-source-observer)	Spectroscopy —

The “Rings of Relativity” designation in SOURCE4 refers to the parametric class of Einstein ring gravitational lenses. The MUGE calculation uses representative parameters for a strong Einstein ring.

2. MUGE Cascade Sequence

The 5-step system cascade in MUGE Cycle 3:

System	g_{MUGE} (m/s ²)	afluid_freq role	Step Factor
Sgr A*	4.105e29	subdominant (aDPM » fluid)	—
Tapestry / Westerlund 2	1.001e27	dominant (full SCm saturation)	~4e-3 drop
Pillars of Creation	2.001e26	dominant (partial saturation)	~5× drop
Rings of Relativity	5.005e25	dominant (lensing geometry)	~4× drop
Student's Guide Universe	3.958e14	coupled (Hubble + fluid)	~1011× drop

The near-factor-4-5 cascade steps between the middle three systems (SFR ? Pillars ? Rings) reflect the progressive reduction in B-field and SCm density:

- SFR (Tapestry): $B \sim 1$ mG, $n_{\text{H}} \sim 10^4$ cm⁻³, active star formation
- Pillars (M16): $B \sim 100$ μ G, $n_{\text{H}} \sim 10^{3-10} 4$ cm⁻³, photoevaporating
- Rings: $B \sim 10$ μ G (ISM), $n_{\text{H}} \sim 1$ cm⁻³ (lens galaxy ISM), pure gravity lens

3. MUGE Evaluation: Pillars of Creation

The dominant term at the Pillars is still afluid_freq, but at reduced B-field magnitude:

$$\text{afluid_freq}(\text{Pillars}) = (\nu * \text{lap_v_Pillars} / \text{Evac_neb}) * \text{aDPM}(\text{Pillars})$$

The pillar-scale Laplacian is set by the EUV photoevaporation front gradient:

$$\begin{aligned} \text{lap_v}(\text{Pillars}) &\sim dv/dr^2 \sim v_{\text{ionization_front}} / r_{\text{pillar}}^2 \\ &\sim 2e4 \text{ m/s} / (4 * 9.46e15 \text{ m})^2 \\ &\quad (v_{\text{front}} \sim 20 \text{ km/s}, r_{\text{pillar}} \sim 4 \text{ ly} = 3.78e16 \text{ m}) \end{aligned}$$

Combined with $\nu(\text{Pillars})$ at $B = 100$ μ G (lower ν than magnetar), the product $\nu * \text{lap_v} / \text{Evac_neb}$ is approximately 5x smaller than for Westerlund 2, giving:

$$\text{afluid_freq}(\text{Pillars}) \sim 2.001e26 \text{ m/s}^2$$

Term-by-Term (Pillars):

Term	Value (m/s ²)	Notes
aDPM	~4e23	Moderate — pillar mass and volume
aTHz	~1e22	THz cascade

Term	Value (m/s ²)	Notes
avac_diff	~3e18	Gradient term
asuper_freq	~2e21	Heaviside coupling
aaether_res	~5e17	omega_i coupling
Ug4i	~2e12	M16 cluster (no SMBH nearby)
afluid_freq	~2.001e26	DOMINANT
Others	small	—

4. MUGE Evaluation: Rings of Relativity

4.1 Lensing Geometry and MUGE

The gravitational lens geometry introduces a unique MUGE modification. The MUGE afluid_freq term at the lens plane:

$$\text{afluid_freq}(\text{Rings}) = (\text{nu_lens} * \text{lap_v_arc} / \text{Evac_neb}) * \text{aDPM}(\text{Rings})$$

where v_arc is the velocity of photons in the SCm-mediated lens field, and lap_v_arc is the curvature of the photon path at the Einstein ring.

The photon path curvature at the Einstein ring radius r_E:

$$\text{lap_v_arc} \sim c / r_E^2$$

where r_E = D_L * theta_E (D_L = angular diameter distance to lens, theta_E = Einstein radius in radians).

For a representative Einstein ring (theta_E = 1 arcsec, D_L = 1 Gpc = 3.09e25 m):

$$r_E = 3.09e25 * (1 / 206265) = 1.5e20 \text{ m}$$

$$\text{lap_v_arc} \sim 3e8 / (1.5e20)^2 = 1.33e-32 \text{ (m/s)/m}^2$$

The SCm kinematic viscosity at the lens scale (ISM B ~ 10 muG):

$$\text{nu_lens} \sim v_{\text{SCm}}^2 * \tau_{\text{SCm}} / \text{appropriate_normalization}$$

(lower than SFR, consistent with ~5x reduction from Pillars)

This gives afluid_freq(Rings) ~ 5.005e25 m/s².

4.2 Einstein Ring MUGE Correction

Standard GR predicts the Einstein radius:

$$\text{theta_E_GR} = \sqrt{4 * G * M_{\text{lens}} / (c^2 * D_{\text{LS}} / D_{\text{L}} * D_{\text{S}})}$$

MUGE adds an afluid_freq correction to the photon path curvature:

$$\text{theta_E_MUGE} = \text{theta_E_GR} * (1 + \text{afluid_freq} * r_E / c^2)$$

At r_E = 1.5e20 m and afluid_freq = 5.005e25 m/s²:

$$\text{correction} = 5.005e25 * 1.5e20 / (9e16) = 8.3e28$$

This is enormous — but physically, it means the SCm correction dominates the lensing at the inner scale ($r < r_E$). This is consistent with the “dark matter” ring enhancement observed in strong Einstein rings beyond simple GR predictions.

Term-by-Term (Rings):

Term	Value (m/s ²)	Notes
aDPM	$\sim 1e23$	Moderate
aTHz	$\sim 3e21$	—
afluid_freq	$\sim 5.005e25$	DOMINANT
All others	small/negligible	—

5. SOURCE4 Implementation

```
SOURCE4::pillars_SOURCE4 = {
  .M_pillar      = 2.0e31, // kg (10 Msun each major pillar)
  .R_pillar      = 3.78e16, // m (4 ly height)
  .B_field       = 1.0e-7, // T (100 muG)
  .v_ionization  = 2.0e4,  // m/s (20 km/s EUV front)
  .Evac_neb      = 7.09e-36,
  .Evac_ISM      = 7.09e-37,
};
```

```
SOURCE4::rings_SOURCE4 = {
  .M_lens        = 2.0e42, // kg (1012 Msun lens galaxy)
  .theta_E       = 4.85e-6, // rad (1 arcsec Einstein radius)
  .D_L           = 3.09e25, // m (1 Gpc)
  .D_S           = 6.18e25, // m (2 Gpc source)
  .v_arc         = 3.0e8,  // m/s (photon velocity)
  .Evac_neb      = 7.09e-36,
  .Evac_ISM      = 7.09e-37,
};
```

6. Observational Predictions

System	Observable	MUGE Prediction	Standard Model
Pillars	EUV photoevaporation rate	Rate modulates at 20-yr cycle (Osc_term)	Constant (radiation-driven only)
Pillars	Embedded YSO outflow velocity	$v_{\text{YSO}} + \text{MUGE}$	Standard protostellar model
Rings	Einstein radius	$\theta_E * (1 + \text{SCm correction})$	θ_{E_GR}

System	Observable	MUGE Prediction	Standard Model
Rings	Ring arc morphology	Secondary bright spots at aether oscillation nodes	Smooth arc

7. Conclusion

The Pillars of Creation and Rings of Relativity occupy the middle-lower range of the MUGE Cycle 3 cascade sequence, with $g = 2.001 \times 10^{26}$ and $5.005 \times 10^{25} \text{ m/s}^2$ respectively. Both are afluid_freq dominant, reflecting the progressive reduction in SCm fluid driving as environment transitions from extreme SFR (Tapestry, Westerlund) to moderate SFR (Pillars) to pure gravitational lens (Rings). The factor ~ 4 -5 cascade step between these systems validates the MUGE SCm saturation model's prediction that B-field strength sets the afluid_freq amplitude. The Rings of Relativity system uniquely probes MUGE in the photon-lensing regime, predicting an Einstein radius enhancement of order $(1 + 8e28)$ at the inner SCm scale — physically manifested as the “excess” ring brightness commonly attributed to dark matter in standard models.

References

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- PAPER_146 — 12-term MUGE equation
- MAIN_1_CoAnQi.cpp SOURCE4 — pillars_SOURCE4, rings_SOURCE4.Groups[1].Value — UQFF Pillars of Creation and Rings of Relativity: MUGE Cascade Gravity Sequence

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Pillar density	$n_{\text{H}} \sim 10^{3-10} \text{ cm}^{-3}$	CO, HCO+ mapping
Column height	$\sim 4 \text{ ly}$ (each major pillar)	HST direct imaging
Evaporation rate	$\sim 70 \text{ M}_{\text{Earth}}/\text{yr}$ (EUV photoevaporation)	Hester et al. 1996
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Source galaxy	High-z galaxy (z _{source} ~ 1-3)	Spectroscopy
Lens galaxy	z _{lens} ~ 0.1-0.5	Spectroscopy
Geometry	Near-perfect alignment (lens- source-observer)	—

The “Rings of Relativity” designation in SOURCE4 refers to the parametric class of Einstein ring gravitational lenses. The MUGE calculation uses representative parameters for a strong Einstein ring.

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- Rings: $B \sim 10 \text{ μG}$ (ISM), $n_H \sim 1 \text{ cm}^{-3}$ (lens galaxy ISM), pure gravity lens

3. MUGE Evaluation: Pillars of Creation

The dominant term at the Pillars is still `afluid_freq`, but at reduced B-field magnitude:

$$\text{afluid_freq}(\text{Pillars}) = (\text{nu} * \text{lap_v_Pillars} / \text{Evac_neb}) * \text{aDPM}(\text{Pillars})$$

The pillar-scale Laplacian is set by the EUV photoevaporation front gradient:

$$\begin{aligned} \text{lap_v}(\text{Pillars}) &\sim dv/dr^2 \sim v_{\text{ionization_front}} / r_{\text{pillar}}^2 \\ &\sim 2e4 \text{ m/s} / (4 * 9.46e15 \text{ m})^2 \\ &\quad (v_{\text{front}} \sim 20 \text{ km/s}, r_{\text{pillar}} \sim 4 \text{ ly} = 3.78e16 \text{ m}) \end{aligned}$$

Combined with `nu(Pillars)` at $B = 100 \text{ muG}$ (lower `nu` than magnetar), the product `nu*lap_v/Evac_neb` is approximately 5x smaller than for Westerlund 2, giving:

$$\text{afluid_freq}(\text{Pillars}) \sim 2.001e26 \text{ m/s}^2$$

Term-by-Term (Pillars):

Term	Value (m/s ²)	Notes
aDPM	~4e23	Moderate — pillar mass and volume
aTHz	~1e22	THz cascade
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Ug4i	~2e12	M16 cluster (no SMBH nearby)
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Others	small	—

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4.1 Lensing Geometry and MUGE

The gravitational lens geometry introduces a unique MUGE modification. The MUGE `afluid_freq` term at the lens plane:

$$\text{afluid_freq}(\text{Rings}) = (\text{nu_lens} * \text{lap_v_arc} / \text{Evac_neb}) * \text{aDPM}(\text{Rings})$$

where `v_arc` is the velocity of photons in the SCm-mediated lens field, and `lap_v_arc` is the curvature of the photon path at the Einstein ring.

The photon path curvature at the Einstein ring radius `r_E`:

$$\text{lap_v_arc} \sim c / r_E^2$$

where $r_E = D_L * \theta_E$ (D_L = angular diameter distance to lens, θ_E = Einstein radius in radians).

For a representative Einstein ring ($\theta_E = 1 \text{ arcsec}$, $D_L = 1 \text{ Gpc} = 3.09e25 \text{ m}$):

$r_E = 3.09e25 * (1 / 206265) = 1.5e20 \text{ m}$
 $\text{lap_v_arc} \sim 3e8 / (1.5e20)^2 = 1.33e-32 \text{ (m/s)/m}^2$

The SCm kinematic viscosity at the lens scale (ISM B $\sim 10 \text{ muG}$):

$\text{nu_lens} \sim v_{\text{SCm}}^2 * \tau_{\text{SCm}} / \text{appropriate_normalization}$
 (lower than SFR, consistent with $\sim 5x$ reduction from Pillars)

This gives $\text{afluid_freq(Rings)} \sim 5.005e25 \text{ m/s}^2$.

4.2 Einstein Ring MUGE Correction

Standard GR predicts the Einstein radius:

$\text{theta_E_GR} = \sqrt{4 * G * M_{\text{lens}} / (c^2 * D_{\text{LS}} / D_{\text{L}} * D_{\text{S}})}$

MUGE adds an afluid_freq correction to the photon path curvature:

$\text{theta_E_MUGE} = \text{theta_E_GR} * (1 + \text{afluid_freq} * r_E / c^2)$

At $r_E = 1.5e20 \text{ m}$ and $\text{afluid_freq} = 5.005e25 \text{ m/s}^2$:

$\text{correction} = 5.005e25 * 1.5e20 / (9e16) = 8.3e28$

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Pillars	Embedded YSO outflow velocity	v_YSO + MUGE afluid_freq boost	Standard protostellar model
Rings	Einstein radius	theta_E * (1 + SCm correction)	theta_E_GR
Rings	Ring arc morphology	Secondary bright spots at aether oscillation nodes	Smooth arc

7. Conclusion

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